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Aircraft parking stand allocation problem with safety consideration for independent hangar maintenance service providers

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ABSTRACT

An aircraft parking stand allocation problem for aircraft hangar maintenance in the context of an independent aircraft maintenance, repair and overhaul (MRO) service provider is studied. This problem arises from the increasing outsourcing maintenance requests initiated by clients that can cause congestion on certain days. Given a set of maintenance requests on a peak day that exceed the capacity of the maintenance hangar, the service provider has to select and first serve the particular subset of aircraft that maximizes their overall profits and then rearrange the remaining requests later. The objective of the proposed problem is to determine a subset of maintenance orders with maximal overall profits and a feasible parking plan on a peak day. In particular, there is to be no overlap between aircraft, and the risk of collision measured by the shortest distance between each pair of aircraft is to be minimized. To solve this problem, No-Fit Polygon (NFP) construction is adopted to prevent overlap between pairs of aircraft. A two-stage MIP approach is proposed, in which the first model is used to find the subset of maintenance orders with the maximal overall profits, while the second model maximizes the overall safety margins based on the revised NFPs. A heuristic algorithm is introduced in order to improve the efficiency of the branch-and-bound algorithm in the second stage problem. Testing instances are generated based on the real situation in an aircraft maintenance company, and the effectiveness of the proposed approaches are evaluated through computational experiments.

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1. Introduction

Aircraft maintenance, repair and overhaul (MRO) activities are critical for the aviation industry. Maintenance checks are carried out periodically to ensure operational safety and reliability. In general, maintenance checks can be categorized into line maintenance and hangar maintenance. The cost of MRO typically represents around 9% of the total annual operating cost for airlines (ITAT 2015), which is the third highest cost center behind the cost of various fuel operations and labor. Operation costs with fluctuation of passenger demand nowadays have forced many airlines to reconsider their business models to ensure that their MRO operations continue to meet the safety requirements while maintaining minimum costs, maximum quality and the best lead-time (Eriksson and Steenhuis, 2014; Knotts, 1999). Outsourcing of MRO operations to an independent company has become an attractive option for airlines, especially heavy maintenance conducted in a hangar. Marcontell (Marcontell, 2013) estimated that the percentage of MRO outsourcing rose 45% from the mid-1990s to 2012.

From the perspective of aircraft hangar maintenance companies (also known as independent MRO service providers), efficiently fulfilling the increasing maintenance requests from clients becomes a challenging task due to limited resource availability. The Operations Manager has to arrange a series of aircraft parking stand allocation plans with a maintenance schedule over a given period of time. In practice, a maintenance hangar fully owned by an airline is predefined into several fixed parking stands at the design stage due to the limited types of aircraft models they own (Fig. 1(a)), and aircraft are then parked at predefined stands during hangar maintenance. Differing from the maintenance hangar owned by airlines in which the capacity of hangar can be easily quantified, independent hangar maintenance companies receive maintenance orders from different clients (e.g. full-cost airlines, low-cost airlines, cargo airlines, government flying services as well as private aircraft owners), which means that the maintenance company needs to flexibly arrange those aircraft in different sizes

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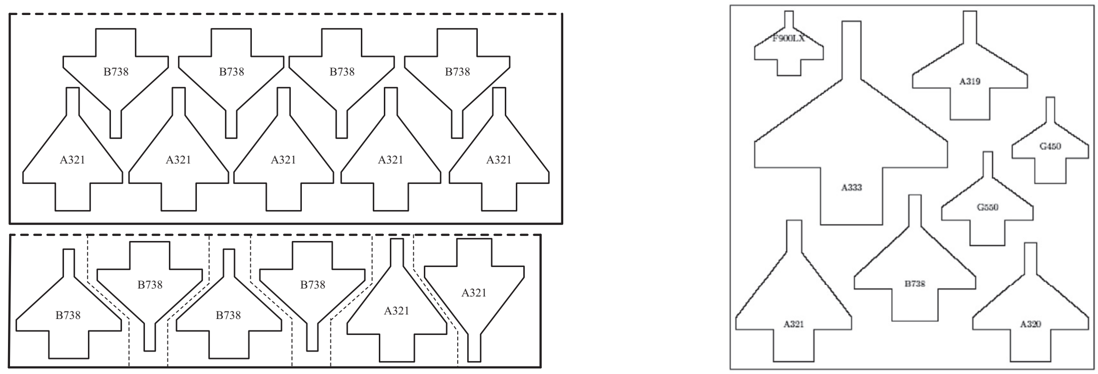


(a) **(b)**

Fig. 1. Airline-owned hangar and independent MRO service provider's hangar.

in the hangar (Fig. 1(b)). Moreover, safety considerations should be taken into account when carrying out an aircraft parking plan.

The proposed parking stand allocation problem arises from the hangar maintenance scheduling problem in the context of aircraft maintenance companies. In practice, each client served by a MRO service provider establishes its own maintenance plan, and then each client proposes its desired maintenance period to the MRO service provider. After receiving the maintenance requests, the maintenance company gathers the subset of aircraft with similar proposed arrival times, service requirements and estimated departure times, and then reviews its maintenance capability and notices their availability to the client. If the maintenance company has the capacity to serve such a set of aircraft, these aircraft are grouped into a "batch" then rolled into the hangar together for maintenance. The maintenance tasks are finished around the same time, and the aircraft depart from the hangar within a similar timeframe. Although maintenance scheduling problems have been extensively studied in the literature, there is no available approach to quantify the capacity of a maintenance hangar, given the variance of the maintenance requests from time to time in the maintenance company, and manual planning is adopted in those maintenance companies that serve aircraft of difference size. Nowadays, the maintenance requests from different customers are likely to cause crowding on some days, and a situation when the hangar cannot accommodate all the incoming aircraft at the same time over the whole planning period is common, given the increasing outsourcing maintenance requests. Under such circumstances, the maintenance company has to negotiate with some clients to adjust their respective maintenance periods. Given that a set of arriving aircraft may exceed the capacity of the hangar space on peak days, the maintenance company aims to come up with a parking plan that makes the most of their hangar space. We are interested in providing an approach that assists independent hangar maintenance companies in maximizing their overall profits while minimizing the risk of collision in the hangar, particularly in situations where the hangar is not capable of accommodating all the incoming maintenance requests from clients during their peak days. The solution provides reference information for the company management in the process of planning the maintenance schedule and in replying to clients' maintenance requests. According to the best knowledge of the authors, no research to date has addressed the hangar space utilization problem in the context of hangar maintenance for independent MRO service providers.

The remainder of this paper is divided into the following sections. In Section 2, we describe the literature related to maintenance planning for aircraft maintenance companies as well as the methodology of preventing aircraft from overlapping. We introduce the concept of the No-Fit Polygon (NFP) that prevents overlap between aircraft in Section 3. In Section 4, the proposed two-stage approach is described in detail. In Section 5, a heuristic algorithm is proposed as a warm start for the exact algorithm, which is followed by discussion on branching strategies. Section 6 presents the results of the numerical experiments and we then analyze the effectiveness of the proposed algorithm. Finally, conclusions are presented in Section 7.

2. Literature review

2.1. Overview of aircraft maintenance and independent MRO service provider

We refer to the classification proposed by Van den Bergh et al. (2013), in which maintenance is categorized into line maintenance and hangar maintenance, according to the place of maintenance. Line maintenance refers to "on line" maintenance that can be conducted when an aircraft parks at the gate or at the apron, and all the other maintenance is categorized as "hangar" maintenance, which needs to be conducted in a maintenance hangar. It is reported that technical delays are the cause of over 20% of disruptions to flight schedules, which is comparable to operational factors (Eriksson and Steenhuis, 2014; Van den Bergh et al., 2013). To minimize the disruption caused by technical problems as well as the cost of maintenance, many airlines have begun to review their MRO operations policy to ensure the airworthiness of each aircraft. Outsourcing of MRO activities is now emerging as a credible solution for airlines, allowing them to focus on their high-value added commercial flying activities. It was estimated that the global commercial aviation MRO market was worth up to USD 60 billion in 2016, and the 10- year demand (from 2016 to 2026) for MRO in business market is estimated to be USD 121.8 billion (Penton's Aviation Week Network 2015).

Given the significant development of independent MRO service companies, few research studies have considered line maintenance problems from the aircraft maintenance company's perspective. Chung et al. (2015) provided an extensive review on proactive maintenance planning issues in the aviation industry. Up to now, some studies have covered workforce scheduling problems from the aircraft maintenance company's perspective. De Bruecker et al. (2015) considered an aircraft maintenance personnel rosters problem in an independent aircraft line maintenance company, which assumed that the maintenance routing problem has been solved within airline companies and the route was given by several airline companies. Liang et al. (2015) considered an aircraft maintenance routing problem in the context of the airline company, and Gavranis and Kozanidis (2015) proposed an exact algorithm to solve a maintenance scheduling problem that maximized the fleet availability of a unit of military aircraft. The problem we consider here is different from the aircraft stand allocation problem in the airport context (Guepet et al., 2015). For airport operations, the aircraft stand allocation problem mainly focuses on allocating aircraft to the parking gate of a terminal or apron near the terminal, for passenger boarding or disembarkation. In such contexts, the accurate physical shapes of aircraft are less critical in scheduling and planning problems as such factors have been incorporated in designing an airport to meet the safety requirements. However, the physical shapes of the aircraft become the major factor influencing the parking plan in the context of maintenance hangar of MRO service providers, since the parking plan changes flexibly from time to time.

2.2. Non-overlapping constraints

Some similarities exist between the aircraft parking stand allocation problem and the cutting and packing problem in a two-dimensional fixed dimension container. The two-dimensional packing problem arises whenever one needs to place irregular items inside a container without overlap. According to Dyckhoff (1990), such irregular shape packing problems are referred to as nesting problems and are mainly characterized by the number of relevant dimensions. A survey conducted by Wäscher et al. (2007) provided an improved topology for cutting and packing problems, and the aircraft parking stand allocation problem can be classified as an output maximization problem in fixed dimensions.

Several approaches have been proposed in the literature to cope with the problems of detecting and preventing overlap between two irregular polygons. The most widely used tool for checking whether two polygons overlap is the No-Fit Polygon (NFP) while other approaches, such as the finite-circle method (Zhang and Zhang, 2009), are also available. Approaches to generate NFP include the orbiting algorithm proposed by Mahadevan (1984) and improved by Burke et al. (2007); the Minkowski sums approach by Milenkovic et al. (Milenkovic et al., 1991), Bennell et al. (Bennell et al., 2001), which was updated by Bennell and Song (2008); and decomposing non-convex polygons into several star-shaped polygons (Daniels et al., 1994) or convex polygons (Agarwal et al., 2002). Bennell and Oliveira (2008, 2009) provided a detailed tutorial on how to generate NFP between two non-convex irregular polygons. Moreover, several integer programming formulations that solve nesting problems have been proposed. Gomes and Oliveira (2006) proposed a Simulated Annealing (SA) algorithm and applied a mixed-integer linear formulation in their items compaction phase. Fischetti and Luzzi (2009) introduced the concept of slice that partitions the region outside the NFP. Alvarez-Valdes et al. (2013) introduced a horizontal slice formulation to enhance the formulation of Fischetti and Luzzi (2009). Martinez-Sykora et al. (2015) adopted horizontal slices in their MIP formulation to solve the irregular pieces packing problem with guillotine cuts. Furthermore, Martinez-Sykora et al. (2017) applied horizontal slices in the formulation of the bin packing problem with free rotations. Recently, Cherri et al. (2016) proposed two robust mixed-integer formulations for the irregular polygon packing problem that decompose the non-convex polygons into several convex pieces in order to generate NFP.

3. No-fit polygon (NFP) construction

3.1. Geometric representation

The geometric shape of the aircraft is characterized as a non-convex polygon (Fig. 2). We denote the reference point of each aircraft to be the middle point at the bottom of the aircraft, and the coordinates of the reference point of aircraft p_i in two-dimensional space are denoted as (x_i,y_i).

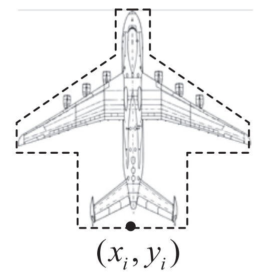


Fig. 2. Geometric representation and reference point of aircraft.

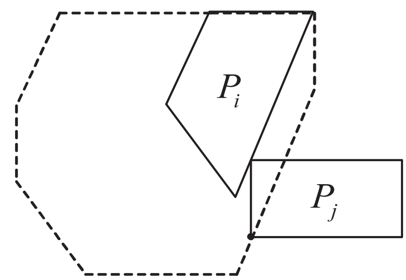


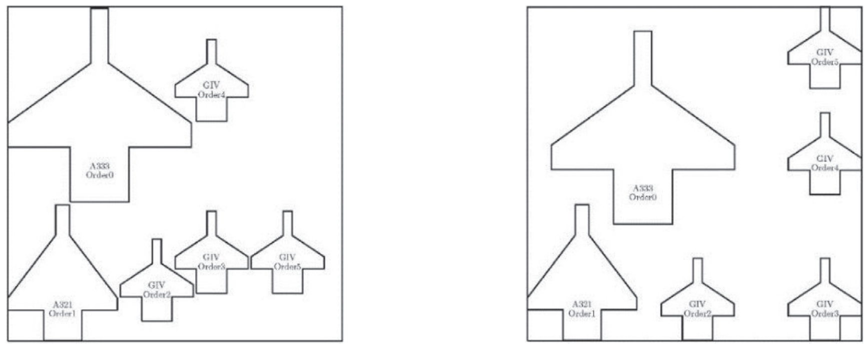
Fig. 3. No fit polygon of P_i and P_j.

3.2. Generation of NFPs

For a pair of polygons p_i and p_j, the No-fit Polygon NFP_{ij} is the region in which the reference point of polygon p_j cannot be placed if polygon p_i remains relative stationary. The feasible zone for placing polygon p_j without overlap with p_i is the region outsideNFP_{ij}. Given these two polygons, the NFP_{ij} is generated by tracing the path of the reference point on p_j as p_j slides around the boundary of p_i such that two polygons always touch but never overlap (Fig. 3). The Minkowski sums approach (Bennell et al., 2001; Bennell and Song, 2008) is a widely used approach generated NFP between two irregular polygons, and is adopted in this paper to generated NFP between aircraft.

4. Mixed integer formulations

The models developed in this section aim to solve the aircraft parking layout allocation problem for independent hangar maintenance service providers. In particular, the objective is to determine a subset of aircraft maintenance requests with maximal overall profits, as well as to finalize the aircraft parking layout plan with maximal overall safety margins. According to the practice adopted in maintenance service providers, as described in Section 1, the aircraft selected to be served during a short period are batched and rolled into the hangar at the same time, with a predetermined sequence before the maintenance tasks begin, then the aircraft are rolled out after all maintenance tasks are finished in the hangar. In this regard, consideration of rolling in and out of a particular aircraft while conducting the maintenance tasks for the other aircraft is not taken into account under such practice, and the assumptions of the problem are as follows: 1) the position of the aircraft in the hangar remains unchanged once rolled into the hangar; 2) for the sequences of rolling in and out, the aircraft in the inner part of the hangar are rolled in first, followed by the subsequent aircraft arranged near the hangar entrance before



(a) Phase I layout

(b) Phase II layout

Fig. 4. Comparison between parking layouts derived from two phases.

maintenance operations; and the roll out operations are conducted in a reverse manner; 3) the maintenance tasks begin after all aircraft are parked in their assigned position, and the aircraft can be rolled out operations until finishing all aircraft maintenance tasks in the hangar. Therefore, the sequence and routing of rolling in and out can be determined according to the outcome of the static parking layout without further stipulation for each case, and the movement operations are not incorporated as a part of the mathematical model for decision making.

We first introduce the concept of Horizontal Slices in Section 4.1, as proposed by Alvarez-Valdes et al. (2013), to prevent overlaps between the polygons in the MIP model. In Phase I, we aim to find the subset of the maintenance order with maximal overall profits. In Phase II, the safety margin of the derived parking plan is considered. It is worth to point out the difference between the problems in these two phases in order to justify the proposed two-stage approach. The objective of the Phase I problem is to find out the subset of aircraft with maximal profits while satisfying the minimal safety margin requirements, therefore, the revised NFPs associated with minimal safety margin are imposed in the model to ensure that aircraft are separated from each other by, at least, the minimal safety distance. The initial parking layout derived from Phase I may be not applicable for practical use. It could be the case in the Phase I solution that aircraft might be placed close to each other or in a concentrated manner in the hangar, even if a large space is left unoccupied and the minimal safety margin is met. In practice, maintenance companies would try to enlarge the distance between pairs of aircraft to minimize the risk of collision during the batch movements, which is different from the cutting and packing problems in the literature that place the items as compactly as possible. The Phase II problem is therefore proposed to provide an exact measurement of the safety margin for aircraft, and finalizes the parking layout. Fig. 4 shows an example of the difference between parking layouts derived from the two phases.

4.1. Concept of horizontal slices

Alvarez-Valdes et al. (2013) improved the approaches of Gomes and Oliveira (2006) and Fischetti and Luzzi (2009) by partitioning the region outside the NFP horizontally (Fig. 5(a)). According to Alvarez-Valdes et al. (2013), each horizontal slice is defined by drawing one or two horizontal line(s) outward from each vertex of the NFP, and they are then characterized by one or two horizontal edge(s) as well as the part of the boundary of the NFP. A set of variables b_{ijk} is associated with each horizontal slice and the reference point of p_j is placed in the slice k if $b_{ijk} = 1$. Therefore, a general form of the constraint preventing overlap is

$$\alpha_{ij}^{kf}(x_j - x_i) + \beta_{ij}^{kf}(y_j - y_i) \le q_{ijk} + M \cdot (1 - b_{ijk}),$$
$$\forall i, j \in P, i \ne j, k = 1, 2, \ldots, m_{ij}$$

where $\alpha_{ij}^{kf}(x_j - x_i) + \beta_{ij}^{kf}(y_j - y_i) = q_{ijk}$ is the equation of the line of the fth edge of the kth slice in NFP_{ij} and m_{ij} is the number of slices outside the NFP_{ij}. To deal with the concavities in NFPs, Alvarez-Valdes et al. (2013) closed them (the dark shade regions associated with variables b_{ij9}, b_{ij10} and b_{ij11} in Fig. 5(a)) until the resulting NFP polygon is convex, then used binary variables to represent each closed concavity. Such treatment is devoted to dealing with the "cave" in the NFP from the vertical direction that is similar to the concavity associated with b_{ij11}in Fig. 5(a), which cannot be represented by the horizontal slices. It is found that such a vertical "cave" does not exist in the NFP between two aircraft and all concavities can be represented by the horizontal slicing method. Therefore, the concavities of the NFPs remain unchanged in our problem and the NFP of two aircraft can be found, as in Fig. 5(b). Generally, the number of binary variables used to label horizontal slices for a NFP between two aircraft is more than 14.

4.2. Phase I

The objective in this phase is to find the subset of maintenance orders with maximal overall profits given a set of maintenance orders that may exceed the capacity of the hangar in fulfilling all of them, while meeting the minimal safety margin between a pair of aircraft. The methodology to enforce the safety margin between aircraft is described in detailed in Section 4.3.1. In general, the profit value of an aircraft is associated with its size, and we prescribe that the profit value for each aircraft is determined by its respective area, since larger aircraft always need more effort in serving and receive more profit, compared with small-sized aircraft. Therefore, the problem is equivalent to minimizing the unused space in the parking plan under such assumption.

4.2.1. MIP formulation for phase I

We first list the parameters, sets and decision variables, along with their associated indices in the Phase I formulation:

Notations

W	width of hangar
H	length of hangar
i	maintenance request associated with aircraft i
P	set of maintenance orders. 1, 2, ..., $i, j \in P$
A_u	a subset of aircraft from$P$ that belongs to the same aircraft type u

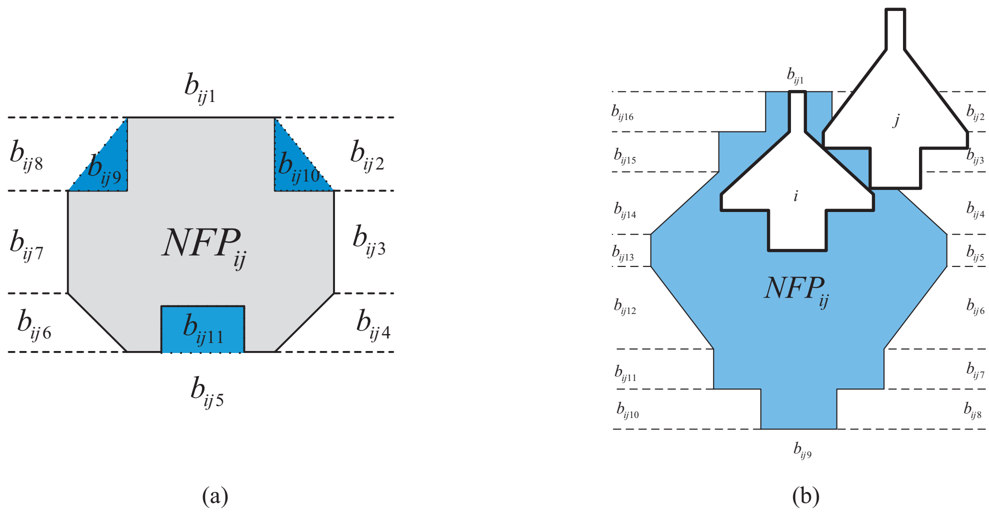


Fig. 5. Horizontal slices outside NFP.

v_i	area of aircraft i
w_i	width of aircraft i
h_i	length of aircraft i
NFP_{ij}	NFP of aircraft i and j
s_{ij}^k	kth slice of the region outside the NFP_{ij}
$\alpha_{ij}^{kf}, \beta_{ij}^{kf}, q_{ij}^{kf}$	parameters used to define the fth linear equation of the slice s_{ij}^k outside the NFP_{ij}
m_{ij}	number of slices outside NFP_{ij}
t_{ij}^k	number of linear equations used to define the slice s_{ij}^k
M	large number

Decision variables

z_i	binary decision variable that takes the value 1 if aircraft i is placed in hangar, and 0 otherwise
x_i	position of reference point of aircraft i on x-axis in two-dimensional space
y_i	position of reference point of aircraft i on y-axis in two-dimensional space
b_{ijk}	binary decision variable that takes the value 1 if the reference point of aircraft j is placed into the slice s_{ij}^k of the region outside NFP_{ij} to separate aircraft i and j, and 0 otherwise

Objective : MaximizeHangarUtilization (F1)

$$Min\ W \cdot H - \sum_{i \in P} v_i z_i \tag{1}$$

s.t.

$$x_i + w_i/2 \le W, \ \ \forall i \in P \tag{2}$$

$$x_i \ge w_i/2 - M \cdot (1 - z_i), \ \ \forall i \in P \tag{3}$$

$$y_i + h_i \le H, \ \ \forall i \in P \tag{4}$$

$$\alpha_{ij}^{kf}(x_j - x_i) + \beta_{ij}^{kf}(y_j - y_i) \le q_{ij}^{kf} + M \cdot (1 - b_{ijk}),$$
$$\forall i, j \in P, i \ne j, \ \ \forall k = 1, 2, \dots, m_{ij}, \ \ \forall f = 1, 2, \dots, t_{ij}^k \tag{5}$$

$$\sum_{k=1}^{m_{ij}} b_{ijk} \le z_i, \ \ \forall i, j \in P, i \ne j \tag{6}$$

$$\sum_{k=1}^{m_{ij}} b_{ijk} \le z_j, \ \ \forall i, j \in P, i \ne j \tag{7}$$

$$\sum_{k=1}^{m_{ij}} b_{ijk} \ge z_i + z_j - 1, \ \ \forall i, j \in P, i \ne j \tag{8}$$

$$z_i \ge z_j, \ \ \forall i, j \in A_u; i < j \tag{9}$$

$$b_{ijk} \in \{0, 1\} \ \forall i, j \in P, i \ne j, k = 1, 2, \dots, m_{ij} \tag{10}$$

$$z_i \in \{0, 1\} \ \forall i \in P \tag{11}$$

$$x_i, y_i \ge 0 \forall i \in P \tag{12}$$

The objective function (1) minimizes the unused space in the maintenance hangar, which is equivalent to maximizing the overall profits of the parking plan. Constraints (2)–(4) are bound constraints that prevent aircraft exceeding the hangar boundary. As the reference point of each aircraft is on the middle point of the bottom, x_i of aircraft i should be equal or larger than $w_i/2$to keep the aircraft within the boundary of the hangar (Constraint (3)). Constraint (5) prevents overlap between each pair of aircraft in the hangar. b_{ijk} are sets of binary variables and one of them must take the value 1 if aircraft i and j are placed in the hangar. Constraints (6)–(8) indicate that the non-overlapping constraints associated with the minimal safety margin requirement between aircraft i and j are activated only if two aircraft are placed in the hangar. For those aircraft to be placed in the hangar, $z_i = 1$ activates constraints (3) and (5)–(8) to impose boundary and non-overlapping constraints for each aircraft in the hangar, otherwise $z_i = 0$ deactivates non-overlapping constraints for those aircraft that are not going to be placed in the hangar. Constraints (10) and (11) indicate that b_{ijk} and z_i are binary variables.

It is common that several incoming maintenance requests have the same type of aircraft and we use A_u to denote those sets of aircraft that belong to aircraft type u. If these aircraft belong to the same aircraft type from different clients are considered as different aircraft in the problem, the algorithm has to fathom partial and complete solutions, which are identical to the other solutions already studied in other branch trees. Therefore, Constraint (9) sets $z_i \ge z_j (i,j \in A_u;\ i < j)$ for the aircraft that belong to the same aircraft type in order to prevents duplicate solutions and saves unnecessary computational effort.

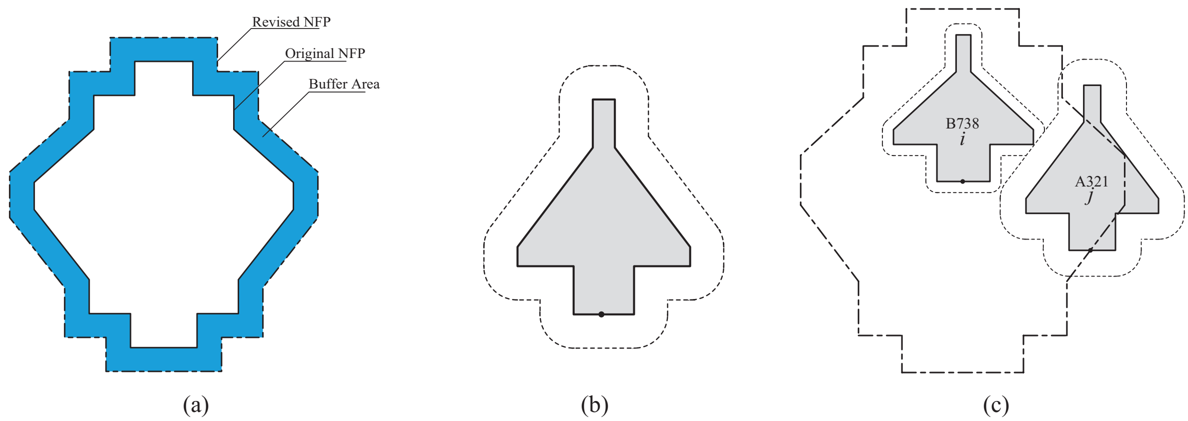


Fig. 6. Revised NFP and safety margin of aircraft.

4.3. Phase II

After the subset of maintenance orders with maximal overall profits is found, the overall safety margins in the hangar are maximized in the Phase II problem. We define the safety margin as the shortest distance between two aircraft in two-dimensional space (Fig. 6(c)) and the overall safety margins are calculated based on the weighted sum of the individual safety margin of each aircraft placed in the hangar. As described at the beginning of Section 4, the proposed models assume that rolling in and out operations are conducted in a batch before conducting maintenance tasks and after finishing all maintenance tasks in the hangar respectively. With this regard, the manoeuvrability and risk of collision is considered to facilitate the batch movements operations, and the rolling in and out sequences are not in incorporated in the decision-making, as mentioned in Section 4.1. Large-sized aircraft bear more risk of collision than medium-sized and small sized aircraft since larger aircraft are not as manoeuvrable as small aircraft during the batch movement operations. Therefore, the large-sized aircraft are given higher priorities so as to reserve larger safety margin in practice, which enhances the smoothness of the batch movement operations. To meet the practical requirement in maintenance operations, we prescribe that the weight of each aircraft in the objective function in the Phase II problem is determined by the respective aircraft area.

4.3.1. Revised NFPs

For a pair of aircraft, the distance used to separate the two aircraft is determined by the aircraft with larger safety margin. In Fig. 6(c), assume that the safety margin of aircraft i (aircraft type: B738) and j (aircraft type: A321) are 3 m and 5 m, respectively. Since aircraft j has a larger safety margin than aircraft i, the shortest distance between the two aircraft is determined by aircraft j. Imposing a safety margin for an aircraft is equivalent to adding a buffer area outside each aircraft in order to prevent other aircraft from entering the buffer area from any direction, and therefore the buffer area outside an aircraft is formed by edges and arcs (Fig. 6(b)). According to the definition of NFP, aircraft j touches aircraft i if the reference point of aircraft j is placed on any edge of NFP between aircraft i and j. Moving the edges of NFP for a pair of aircraft outward is equivalent to enlarging the boundary of non-allowable area for the reference point of the relative movable aircraft in that pair, i.e. aircraft j in Fig. 6(c). Therefore, to impose a safety margin n between two aircraft, we revise the original NFP to separate the two aircraft by moving each edge of the original NFPs outward by distance n (Fig. 6(a) and (c)).

We discretize the safety margin of the aircraft and prescribe the lower bound lb and upper bound ub of the safety margin n ($lb \leq n \leq ub$) in consideration of practical applications and the tradeoff between the accuracy of the parking plan and the computational efficiency. In actual operation, aircraft are given the coordinates of the parking stands then towed or pushed to the assigned position by a tow truck. Pushing or towing aircraft to the exact position assigned by the planner is rather difficult in reality, and therefore a deviation tolerance between the actual and assigned positions is usually given. Therefore, setting a safety margin as a continuous variable is intended to increase the accuracy of the parking layout, however such an assumption is not that practical in the actual situations. In addition, setting safety margins as continuous also increases the computational burdens on the Phase II problem: we have to construct a significant number of revised NFPs with a set of binary variables. Similarly, the upper bound of the safety margin provides an interpretation in the context of practical operation: separating two aircraft with a safety margin larger than the upper bound of the safety margin does not contribute to the overall safety margin and therefore is not considered in decision-making. Such a limit also controls the scale of the problem in Phase II. As we revise the original NFP, i.e. enlarge the original NFP by removing each edge of the original NFP outwards so as to construct a set of revised NFPs to separate aircraft by a given safety margin n in Phase II, activating the revised NFP with safety margin n indicates that the algorithm tries to impose the shortest distance between this pair, at least equal to n, or larger than n.

4.3.2. MIP model for Phase II problem

The formulation of the Phase II problem (F2) is similar to that of the Phase I problem (F1) with the modification on activation/deactivation of non-overlapping constraints. The primal decision variables of the Phase II problem are the individual safety margins z_i^n of the subset of aircraft $i \in P'$ that are derived from the Phase I problem with maximal overall profits. As seen in Fig. 4, the aircraft parking layout is finalized by enlarging the safety margin of each aircraft in the Phase II problem, with an objective function $Max\ \sum_i^{P'} \cdot \sum_{n=lb}^{ub} z_i^n \cdot n \cdot Area_i$ where z_i^n denotes the binary decision variable associated with admissible safety margin n ($lb \leq n \leq ub$) of aircraft $i \in P'$ and $Area_i$ refers to the area of the aircraft. NFP_{ij}^n denotes the revised NFPs for a pair of aircraft i and j with safety margin n. Similar to the notation characterizing the non-overlapping constraints in Phase I model, s_{ijn}^k refers to the kth slice outside the region of the revised NFP of aircraft i and j with safety margin n;α_{ijn}^{kf}, β_{ijn}^{kf}, q_{ijn}^{kf} are the parameters used to define the fth linear inequality of the slice s_{ijn}^k outside NFP_{ij}^n; m_{ij}^n is the number of horizonal slices outside NFP_{ij}^n and t_{ijn}^k records the number of linear inequalities of the kth slice outside NFP_{ij}^n. b_{ijk}^n is binary decision variable that takes the value 1 if the reference point of aircraft j is placed into slice s^k_{ijn}, and g^n_{ij} is introduced as auxiliary binary decision variable that takes the value 1 if the shortest distance between aircraft i and j is n and NFP^n_{ij} is activated. The coordinates of the aircraft i, i.e. $x_i$and y_i, are confined by the decision variables mentioned above to finalize their parking positions. Moreover, $A_u$records a group of aircraft that belong to the same aircraft type u, which is used to created constraints preventing duplicate solutions. The constraints involved in the Phase II model consist of: 1) boundary constraints preventing aircraft from exceeding the boundary of hangar; 2) non-overlapping constraints activated/deactivated by b^n_{ijk}, g^n_{ij} and z^n_i; 3) logical constraints prescribing that each aircraft $\forall i \in P'$ must select a safety margin within the admissible range $\sum_{n=lb}^{ub} z^n_i = 1$, and each pair of aircraft is separated by one revised NFP within the admissible range as well, i.e. $\sum_{n=lb}^{ub} g^n_{ij} = 1, \forall i, j \in P', i \neq j$; and 4) the constraint preventing duplicate solutions for the set of aircraft A_u belong to same type u, which prescribes that safety margins for the aircraft belong to the same aircraft type are in a decreasing order, i.e. $\text{margin}_{i1} \geq \text{margin}_{i2} \geq \cdots \geq \text{margin}_{in}$. It may happen that the hangar cannot accommodate the whole subset of aircraft derived from Phase I after imposing safety margins in (F2), if we use the original NFP to solve (F1). To solve this problem, the original NFPs are revised by the lower bound of the safety margins while solving the Phase I problem to impose the minimal safety margin requirement.

5. Branch and bound algorithm

The formulations described in Sections 4.1 and 4.2 can be solved by a branch-and-bound algorithm. In this section, we focus on improving the efficiency of the branch-and-bound algorithm for the Phase II problem by providing an initial solution from the heuristic as well as introducing branching strategies.

5.1. Heuristic algorithm in Phase II problem

Determining the feasibility of a solution in Phase II can be difficult. Specifically, the branch-and-bound has to fathom every possible node of b^n_{ijk} in a given combination of safety margins (set of z^n_i) to prove infeasibility. A heuristic that tightens the upper bound in the Phase II problem (F2) as well as providing a moderate upper bound of safety margin is proposed. Each iteration of the heuristic examines the feasibility of a given combination of safety margins by checking whether the algorithm can feasibly park all aircraft using a MIP model: We construct a MIP model *feasibility check* by revising the formulation of the Phase I problem, i.e. (1)–(12). In detail, the original NFPs are replaced with the revised NFPs determined by the set of safety margins, and the objective function of the *feasibility check* is $\max \sum_{i \in P'} z_i$, which indicates the number of aircraft placed in hangar. If the optimal objective value of the *feasibility check* model equals the number of aircraft in set P', then the hangar is able to accommodate all the aircraft and such a solution is “feasible” in (F2). Otherwise the optimal value is less than the number of aircraft in set P', which implies that at least one aircraft cannot be placed in the hangar with the given safety margin, and such solution is “infeasible” in (F2). The heuristic algorithm can be described in three steps:

Step 1: We try to place the subset of aircraft derived from Phase I in the hangar with some combinations of safety margins (set of $z^n_i, i \in P'$) by calling the *feasibility check*. The initial individual safety margin for each aircraft equals to the lower bound of the safety margin in this step. If all of those aircraft can be feasibly placed in the hangar with the given safety margins, the respective objective value $Max\ \sum_i^{P'} \cdot \sum_{n=lb}^{ub} z^n_i \cdot n \cdot Area_i$ determined by the heuristic solution can be updated as the lower bound in (F2). After that, the safety margins of all those aircraft are augmented and the *feasibility check* is called again until the infeasible solution returns with a value of safety margin, and this value of safety margin is recorded as the threshold value of the safety margin.

Step 2: When a feasible parking plan cannot be found with the given combination of safety margins, the safety margins of some aircraft must be reduced in order to obtain a feasible parking plan. Two strategies of adjusting the safety margin are introduced during the individual safety margin decrementation and augmentation stage, respectively. In particular, we first assign priorities to each maintenance order according to the area of each aircraft and put all aircraft into a set named *Prioirty_List*, and the priorities are assigned in a decreasing order according the area of each aircraft, i.e. assigning the highest priority to the largest aircraft. The first adjusting strategy is that the safety margin for the aircraft with the lowest priority the *Priority_List* is decreased at first until the *feasibility check* can produce a feasible parking solution, or the aircraft is moved to the waiting list set, i.e. *Aug_List*, for further consideration of the safety margin augmentation later if it reaches the lower bound of the safety margin, and the algorithm selects another aircraft with the lowest priority in *Priority_List* for safety margin decrementation. After finding a feasible solution by decrementing the safety margin of the aircraft, the algorithm selects the aircraft with the highest priority for safety margin augmentation from *Priority_List* again and checks the feasibility. If the infeasible solution returns, then such aircraft is removed from further augmentation consideration. The algorithm continues to select aircraft with highest priority for augmentation until the *Priority_List* is empty.

Step 3: After the iterations in **Step 2**, we again obtain a set of aircraft pending for safety margin augmentation in the waiting list *Aug_List*, and the second adjusting strategy is proposed. In detail, the safety margin for aircraft with higher priorities in the waiting list *Aug_List* are augmented first, then aircraft with lower priorities are augmented later until the *feasibility check* returns infeasible solutions and such aircraft causing infeasibility in this iteration is removed from *Aug_List* for further consideration of safety margin augmentation.

5.2. Branching strategies

It is noted that the binary decision variables in both problems have hierarchical structures, and the branching strategies can be adjusted to adopt this feature. Fischetti and Luzzi (2009) proposed a set of branching strategies in open dimension nesting problems and Alvarez-Valdes et al. (2013) conducted an extensive study to compare the efficiency among various branching strategies. The strategy proposed by Fischetti and Luzzi (2009) is first to determine the relative positions of 2 pieces, then those of 3 pieces, then 4 pieces, and so on, by assigning priorities to the variables b_{ijk} in a decreasing order. This strategy avoids visiting the subtrees that produce infeasible solutions before the search process. Similarly, in the Phase I problem, a set of binary variables z_i is used to first decide which aircraft is going to be placed in the hangar, and then set b_{ijk} for each pair of aircraft that is branched to find the relative position for those aircraft with $z_i = 1$. According to Alvarez-Valdes et al. (2013), assigning a higher priority to a larger piece first yields satisfactory results. Similar to their approach, we first assign priority to the variable z_i using Fischetti and Luzzi (2009)'s priorities by ordering pieces by the non-increasing area in (F1), and then we assign priority to variables b_{ijk} that are used to separate each pair of aircraft. Similarly, in the Phase II problem, the decision variables

Table 1
Frequency of daily handling maintenance requests.

Number of aircraft to be arranged in one day	Frequencies	Percentage (%)
0	1	0.6
1	11	7
2	21	13.4
3	23	14.6
4	24	15.3
5	23	14.6
6	19	12.1
7	18	11.5
8	10	6.4
9	7	4.5

Table 2
Classification of aircraft models.

Category	Aircraft models
Small-sized aircraft (aircraft area $<300\ m^2$)	G200 CL600 CL605 F900LX F2000EX F2000LX ERJ135 F7X G450 GIV
Medium-sized aircraft ($1500\ m^2 >$ aircraft area $>=300\ m^2$)	GL5T G550 G5000 G6000 G650 A318 ERJ190 A319 A320 B738 A321
Large-sized aircraft (aircraft area $>=1500\ m^2$)	A332 A333

that determine the safety margin for each aircraft z_i^n are branched first, and auxiliary variables g_{ij}^n are determined afterwards. Thirdly, binary variables b_{ijk}^n are branched under given sets of z_i^n and g_{ij}^n.

6. Numerical study

This section presents the results of computational experiments that were carried out on instances based on the real-life data provided by an aircraft maintenance company in Hong Kong. All the procedures described in the previous sections are coded in C# in Visual Studio 2010 and run on a computer with an Intel Core i7 processor, at 3.6 GHz with 32 Gb of RAM. The Mixed-Integer Linear Programming is solved by the CPLEX 12.5 serial model.

6.1. Description of instances

An aircraft hangar maintenance service provider in Hong Kong, which has over 50 clients, including airlines, business jet companies and utility aircraft companies, is employed as a case study. The company owns a maintenance hangar in the aircraft maintenance area of Hong Kong International Airport, with dimensions are at 110 m long and 110 m wide. In the current maintenance scheduling process of the company, the parking stand plans are constructed manually at regular periods. We collected the information of the estimated arrival time (ETA), departure time (ETD), aircraft type and check type of each maintenance order from clients over 157 days from January to May in 2015 (see Table C1 in Supplementary Material C for detailed description of the data). We calculated the number of aircraft to be arranged each day by the ETA and ETD of each maintenance order, and the frequencies of the days and number of aircraft to be arranged are presented in Table 1. Typically, the maintenance company has to arrange a mix of large-, middle-, and small-sized aircraft in the hangar each day, and the company management stated that planning for 7 aircraft simultaneously by the manual method is challenging. The data in Table 1 suggests that over one fifth of the days in these five months are peak days for the maintenance planning; it is estimated that the proportion of peak days is going to increase in the future as outsourcing MRO activities increases. 40 testing instances are generated based on the observed peak-day scenarios and the number of aircraft maintenance orders in those instances ranges from 6 to 12 (See Table B1 in Supplementary Material B for a detailed description of the problem instances). In detail, the observed peak-day scenarios are incorporated as initial instances in the set at first, then those instances are altered by adjusting the proportion of small-, medium- and large-sized aircraft as well as adding new aircraft in order to create challenging problem instances.

The instance set consists of 10 small-sized, 11 medium-sized and 2 large-sized aircraft types, including large-sized civil aircraft, medium-size civil aircraft as well as business jets and the classification of aircraft models is based on their area, with Table 2 showing details of the classification (See Fig. A1 and Table A1 in Supplementary Material A for a detailed description of the dimension parameters of aircraft). 40 instances are classified into four groups by the proportion of aircraft from the three categories, as follows: 1) the majority of aircraft in the instance are small-sized, 2) the majority of aircraft in the instance are medium-sized, 3) the majority of aircraft in the instance are large-sized and 4) the number of aircraft from different categories in the instance are equal. With regard to the safety margin range, we referred to the practice in the company and prescribed the minimal distance between two aircraft as one meter and the maximal safety margin as eight meters. To avoid the situation in which the hangar cannot accommodate the whole subset of aircraft derived from Phase I after imposing the safety margin in the Phase II problem, all the original NFPs in the Phase I problem are revised by the one-meter safety margin, which accords with the minimum safety requirement between pairs of aircraft adopted in practice.

6.2. Evaluating the models' performance

6.2.1. Computational results- Phase I

Table 3 shows the computational results for 40 instances in the Phase I problem. The first column stands for the instance name and the "total_S/M/L" in the second column stands for the total number of aircraft to be arranged in such instance and number of aircraft from the three categories, respectively. The number of binary variables involved in each instance is indicated in the third column. The best objective incumbent values in the Phase I problem is indicated in the fourth column. The column ('CPU) and ('Gap') in the fifth and sixth column report the CPU time elapsed when the termination criterion was met as well as the relative gap, respectively. The seventh column represents the number of aircraft placed in the hangar after the stopping criterion was met and the proportion of used hangar space is presented in the last column. The time limit for each instance was 3600 s.

The branch-and-bound algorithm with the revised branching strategy was able to optimally solve 34 instances out of 40. It was found that the highest utilization is 0.54 in the 40 instances and the generated layouts demonstrate packed hangars for the instance with high utilization. There were 6 instances, i.e. 20, 30, 37, 38, 39 and 40, that were left unsolved to optimal. The generated layouts of these 6 instances were further examined and it was found that the hangar space was almost fully utilized with a compact layout. We further identified the aircraft left unarranged in these 6 instances and found that the majority of unarranged aircraft, i.e. pending aircraft, were large-sized and middle-sized aircraft, and it would be unlikely to park such unarranged aircraft, given the existing layouts. Therefore, the interpretation of the optimality gap in these 6 instances can be provided: the difference between the best-known integer solution and the lower bound corresponds to the area of the subset of the pending aircraft and the gap is updated whenever a pending aircraft is successfully placed in the existing layout or it is proven that one of the pending aircraft cannot be placed in the existing layout. Due to the non-convex shape of aircraft, the number of binary variables involved in each NFP between two aircraft is large compared with the convex shape polygons in the problem instances of the other nesting problems

Table 3
Computational results of the Phase I problem.

Instance	Aircraft total_S/M/L	Binary variables	Objective Function	CPU	Gap	Number of aircraft placed in hangar	Proportion of used hangar space
1	6_4/1/1	400	8430.26	0.46	0	6	0.29
2	6_3/2/1	396	8497.08	0.19	0	6	0.30
3	7_4/1/2	543	7711.13	0.12	0	7	0.35
4	7_3/2/2	583	7261.4	0.19	0	7	0.39
5	8_5/2/1	688	6931.33	0.10	0	8	0.32
6	8_4/2/2	738	6770.9	1.02	0	8	0.44
7	9_7/1/1	993	7884.61	0.45	0	9	0.34
8	10_8/1/1	1236	7461.56	0.54	0	10	0.38
9	11_9/1/1	1469	7409.36	0.93	0	11	0.33
10	12_8/3/1	1644	5333.27	128.06	0	12	0.46
11	6_1/3/2	384	6665.39	0.78	0	6	0.45
12	6_0/4/2	426	7456.76	0.47	0	6	0.38
13	7_0/5/2	567	6251.38	0.98	0	7	0.47
14	8_1/5/2	698	5441.4	904.19	0	7	0.51
15	8_2/4/2	704	6929.64	82.56	0	8	0.45
16	8_1/5/2	734	5965.05	3.55	0	8	0.49
17	9_2/5/2	969	6941.82	1.17	0	9	0.44
18	10_3/7/0	1192	8717.2	0.53	0	10	0.29
19	11_1/9/1	1435	7253.9	115.84	0	10	0.45
20	12_2/9/1	1620	6545.54	3600	13.45	10	0.48
21	4_1/1/2	162	7846.49	0.16	0	4	0.33
22	6_1/2/3	388	6279.82	17.93	0	5	0.44
23	6_2/1/3	392	6315.32	0.28	0	6	0.43
24	7_2/1/4	587	6253.1	17.76	0	6	0.44
25	7_2/2/3	575	6232.82	0.47	0	6	0.44
26	7_1/2/4	384	6697	50.06	0	5	0.44
27	8_2/2/4	712	6356.49	112.86	0	6	0.46
28	10_3/3/4	1220	5646.73	469.26	0	8	0.50
29	10_3/2/5	1234	5698.1	710.76	0	8	0.48
30	12_2/5/5	1784	4972.67	3600	48.14	9	0.54
31	6_2/2/2	386	8031.06	0.38	0	6	0.36
32	6_2/2/2	414	7473.4	0.39	0	6	0.34
33	6_2/2/2	392	6969.43	0.38	0	6	0.41
34	9_3/3/3	945	5926.93	554.89	0	8	0.50
35	9_3/3/3	971	5565.00	19.61	0	9	0.51
36	9_3/3/3	939	5893.23	2.61	0	8	0.49
37	12_4/4/4	1776	4669.42	3600	36.49	9	0.50
38	12_4/4/4	1742	5641.33	3600	35.30	8	0.45
39	12_4/4/4	1692	5727.48	3600	39.09	8	0.39
40	12_4/4/4	1764	5196.22	3600	33.65	10	0.53

in the literature (Alvarez-Valdes et al., 2013; Cherri et al., 2016; Toledo et al., 2013). Visiting all the possible relative positions for each pair of aircraft before updating the bound is difficult, even the generated layout already demonstrates a satisfactory result for hangar space utilization.

6.2.2. Computational results of the Phase II problem

The number of binary variables used to determine the relative positions between each pair of aircraft in the Phase II problem is determined by two factors: 1) the number of aircraft derived from the Phase I problem, and 2) the range of the safety margin in the Phase II problem. It is reported that the branch-and-bound algorithm with the *horizontal slicing* MIP model can optimally solve the open-dimensioned nesting problem with at most 14 pieces, with convex- and non-convex shaped pieces (Alvarez-Valdes et al., 2013). To control the problem size to a moderate level, the upper bound of safety margin is determined by the threshold value derived from the heuristic in this section, and the results are shown in the last column in Table 4. Considering that the Phase II problem is similar to the nesting problem with *piece* $\bullet$ $(ub - lb + 1)$ pieces and such a problem size is challenging, we referred to the stopping criterion adopted in the literature. Alvarez-Valdes et al. (2013) set several time milestones (1 h, 2 h, 5 h and 10 h) in solving difficult instances that involved more than 12 complex irregular items. We therefore selected and prescribed the time limit for each instance in Phase II as 18,000 s (5 h) with the upper bound of the safety margin determined by the threshold value derived from the heuristic.

Table 4 shows the computational results in the Phase II problem. We found that in many instances, the initial solution provided by the heuristic was proved optimal by the exact algorithm after an exhaustive search in some large instances. Although, in some cases, the objective function value increased in the branch-and-bound algorithm, the searching processes take a long time. In the Phase II problem, the difference between the best integer and upper bound corresponds to the pending safety margin of each aircraft. To update the upper bound of branch-and-bound, one has to prove the infeasibility of the pending safety margins, which corresponds to a single *feasibility check*. Though the final layout demonstrates a satisfactory result from the practice perspective, large gaps have been recorded in instances with packed layouts since updating the bounds in the Phase II problem is far more difficult than that of the Phase I problem, as there is more than one NPF used to separate the aircraft.

Moreover, the performance of the model and the computation time differs, even when comparing two instances from the same instance group with the same number of aircraft to be arranged in the Phase II problem. Taking Instances 23 and 24 as examples, both have 6 aircraft to be arranged in these two instances in the Phase II problem (as determined by the result of the Phase I problem in Section 6.2.1) and they belong to the same instance group. Moreover, the threshold values derived from the heuristic are the same, and the number of binary variables involved in the two instances are similar with similar instance settings. The exact algorithm takes much more time to solve instance 23 to optimal than for instance 24.

Table 4
Computational results of the Phase II problem.

Instance	Aircraft	Binary variables	CPLEX + Threshold				Heuristic		
	total_S/M/L		LB	UB	CPU	Gap	CPU	Value	Threshold
1	6_4/1/1	2792	29,357.92	29,357.92	0.39	0	1.67	29,357.92	8
2	6_3/2/1	2732	28,823.36	28,823.36	0.30	0	1.37	28,823.36	8
3	7_4/1/2	3822	35,110.96	35,110.96	0.53	0	3.45	35,110.96	8
4	7_3/2/2	4038	38,709.12	38,709.12	8.94	0	2.41	38,709.10	8
5	8_5/2/1	4832	41,349.36	41,349.36	0.78	0	4.32	41,349.36	8
6	8_4/2/2	4498	34,188.06	37,303.56	18,000	9.11	439.05	32,945.88	7
7	9_7/1/1	6888	33,723.12	33,723.12	99.20	0	5.05	33,723.12	8
8	10_8/1/1	8508	31,560.05	37,107.52	18,000	17.58	2314.18	36,065.52	8
9	11_9/1/1	10,224	30,259.88	37,525.12	18,000	24.01	1252.40	35,674.76	8
10	12_8/3/1	7172	N/A	54,133.84	18,000	N/A	5398.84	27,066.92	5
11	6_1/3/2	1356	17,756.84	17,888.13	1689.97	0.74	85.17	17,756.84	4
12	6_0/4/2	2944	35,671.30	35,671.3	2374.99	0	43.08	34,961.92	8
13	7_0/5/2	1984	22,436.38	22,436.38	11,012.39	0	181.61	22,366.38	4
14	8_1/5/2	920	N/A	10,995.12	18,000	N/A	208.63	7643.21	2
15	8_2/4/2	3688	25,741.26	31,022.16	18,000	20.52	401.31	27,928.82	6
16	8_1/5/2	1942	17,378.80	18,404.85	18,000	5.90	1548.54	15,513.60	3
17	9_2/5/2	5100	27,463.44	30,949.08	18,000	12.69	2466.59	28,225.34	6
18	10_3/7/0	8286	27,062.56	27,062.56	546.80	0	20.80	27,062.56	8
19	11_1/9/1	4120	N/A	20,296.40	18,000	N/A	697.94	19,725.02	4
20	12_2/9/1	1966	N/A	11,108.92	18,000	N/A	371.91	10,634.96	2
21	4_1/1/2	1136	34,028.08	34,028.08	0.14	0	0.80	34,028.08	8
22	6_1/2/3	741	12,882.54	12,882.54	35.71	0	15.35	11,640.36	3
23	6_2/1/3	1041	12,950.94	12,950.94	2672.73	0	80.37	11,716.44	3
24	7_2/1/4	1107	12,962.70	12,962.70	1500.32	0	927	11,693.80	3
25	7_2/2/3	1101	13,023.54	13,023.54	2616.64	0	49.60	11,734.36	3
26	7_1/2/4	486	5564.0	5564.0	95.91	0	3.08	5564.40	2
27	8_2/2/4	1065	12,652.53	12,652.53	471.12	0	55.01	11,487.02	3
28	10_3/3/4	2058	12,783.41	19,359.81	18,000	51.44	1195.67	12,906.54	3
29	10_3/2/5	1998	14,263.70	19,205.70	18,000	34.65	353.64	12,803.80	3
30	12_2/5/5	1730	N/A	14,254.66	18,000	N/A	2563.09	8669.56	2
31	6_2/2/2	2700	34,444.88	34,444.88	0.95	0	1.64	32,551.52	8
32	6_2/2/2	2876	37,012.80	37,012.80	0.33	0	1.67	37,012.80	8
33	6_2/2/2	2736	39,290.44	39,290.44	739.98	0	168.87	39,290.44	8
34	9_3/3/3	1922	16,592.67	18,519.21	18,000	11.61	1888.20	16,592.67	3
35	9_3/3/3	1348	10,025.78	13,392.80	18,000	33.58	1271.06	10,025.78	2
36	9_3/3/3	1264	N/A	12,858.98	18,000	N/A	662.18	9080.56	2
37	12_4/4/4	2114	N/A	14,861.16	18,000	N/A	2249.97	11,141.74	2
38	12_4/4/4	1670	N/A	51,016.56	18,000	N/A	1102.94	12,143.07	2
39	12_4/4/4	1672	N/A	12,745.04	18,000	N/A	1509.96	9553.34	2
40	12_4/4/4	2114	N/A	13,897.12	18,000	N/A	2008.41	10,449.86	2

6.3. Discussion

The hierarchical structures of binary variables in the two formulations are analysed, and a branching priority to each binary variable is assigned, which is based on the strategies proposed by Fischetti and Luzzi (2009). Moreover, the range of the admissible safety margin is another factor that influences the computational performance of the Phase II problem. For the tradeoff between the solution quality and the computational efficiency, the solution obtained by the heuristic can be considered for practical use, as the accuracy of the safety margins has been significantly improved compared with the manual layout planning method. From a theoretical point of view, some alternate formulations, such as the *semi-continuous* model and the *discrete* (*dotted-board formulation*) model (Toledo et al., 2013; Leao et al., 2016) can be considered to in future studies simplify the geometric complexity of aircraft.

7. Conclusions

To assist independent MRO service providers in planning the parking stand allocation plan and in reviewing the hangar capacity, especially on peak days, we propose a two-stage approach based on MIP models to solve the problem. The outlined aircraft parking stand allocation problem offers several areas for further work. Concerning hangar space utilization, arranging the aircraft parking stand in a three-dimension space can be considered. For example, small aircraft can be placed under the wings of large aircraft in the real situation. In addition, the analysis of performance has shown that there is still room for improvement in the problem-solving approaches. Given the NP-hardness of this problem, further avenues for future work are the inclusion of additional cut and branching strategies and pruning techniques in the branch-and-bound, as well as the development of efficient metaheuristics. The models we addressed in this work solve a single peak day parking stand allocation problem. It is possible to incorporate the proposed problem into the maintenance scheduling problem in the context of aircraft maintenance company so as to fill the research gap. In this regard, potential future research for this problem is incorporating the parking stand allocation problem aimed at reviewing single day capacity into the maintenance scheduling problem over the entire planning period, which improves the current practice adopted in maintenance companies.

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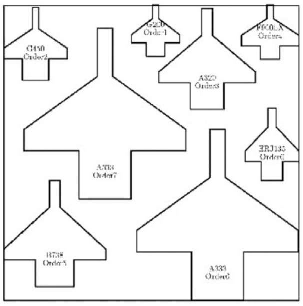
Appendix: Selected best known parking layouts for 20 instances in Section 6.2.2



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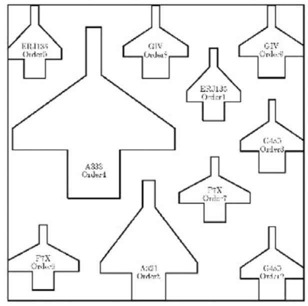

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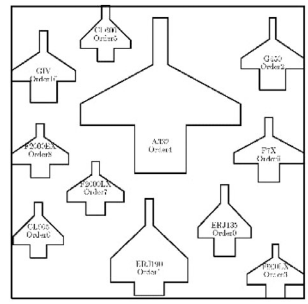

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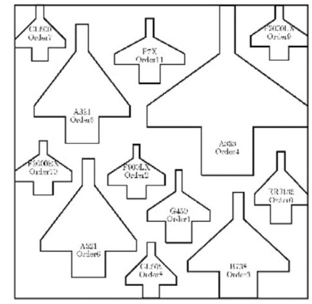

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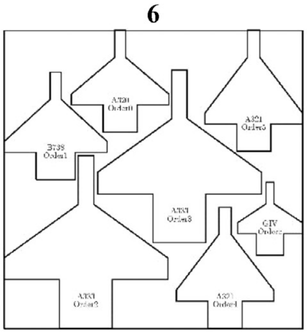

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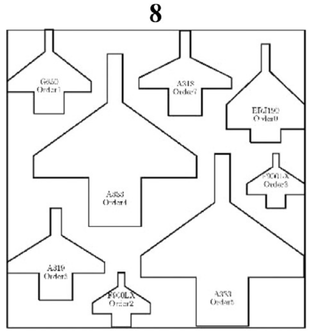

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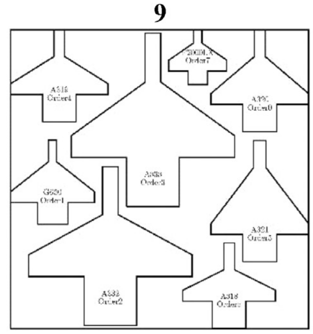

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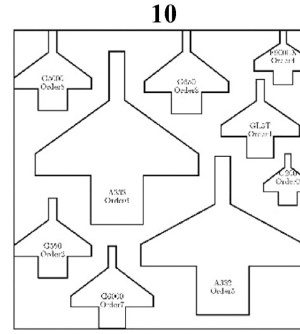

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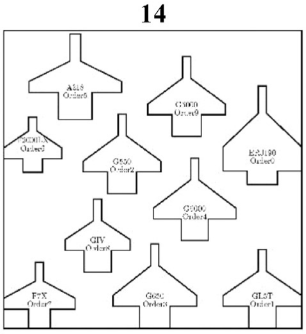

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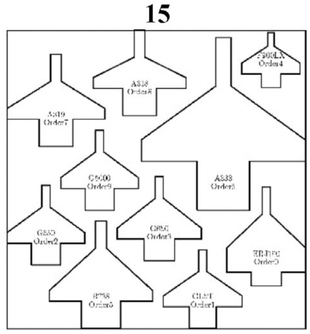

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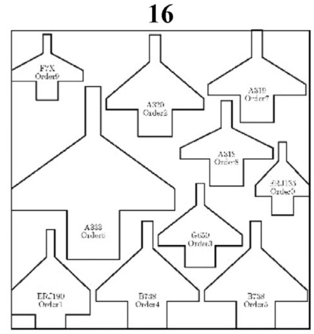

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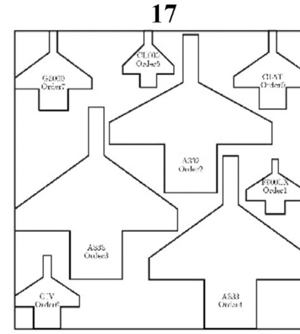

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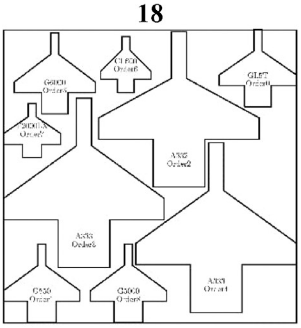

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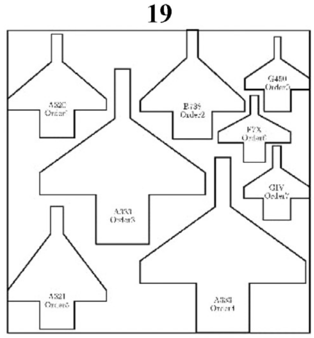

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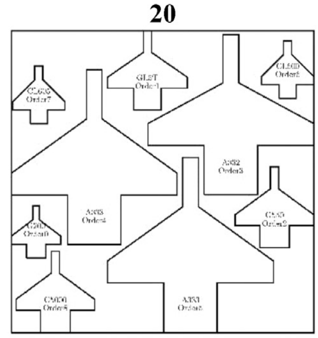

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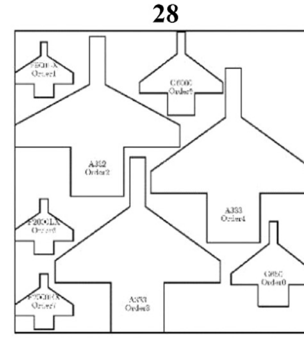

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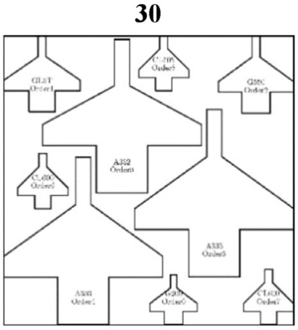

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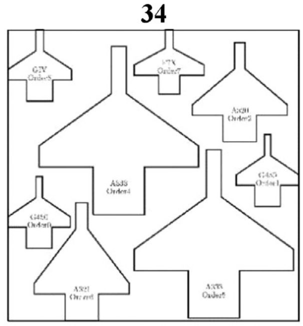

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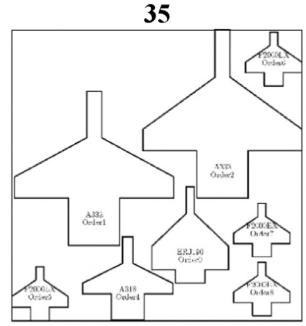

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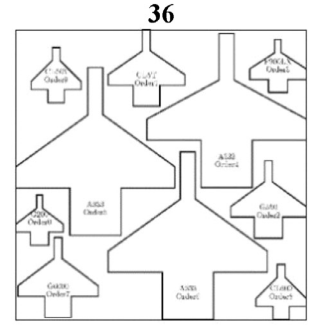

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cor.2017.10.001.

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